A negative fiber-dimension result for Raynaud's $\mathcal{BL}_1 L_q$ axiomatizability problem

Automatic Math Discovery Run `fa_banach_001`

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Status

This packet records a partial negative result related to Raynaud's final question in [1]. It does not answer the unrestricted question whether $\mathcal{BL}_1 L_q$ is axiomatizable as a Banach-space class for $1 < q \leq 2$. It proves that the natural fiber-dimension-restricted variants $\mathcal{BL}_1 L_q^{\geq n}$, $n \geq 2$, are not axiomatizable in that range.

Source question

Raynaud summarizes the known positive cases for the Banach-space axiomatizability of mixed-norm classes and then asks:

> space, see [2[illegible]]. □
>
> The main question remaining open in the context of $\mathcal{B}L_p L_q$-spaces is:
>
> *Question* 6.17. Is the class $\mathcal{B}L_1 L_q$ axiomatizable in the language of Banach spaces, for $1 < q \leq 2$?
>
> REFERENCES

Question 6.17 of [1] asks whether

$$\mathcal{BL}_1 L_q$$

is axiomatizable in the language of Banach spaces for $1 < q \leq 2$. Earlier in the same section Raynaud proves positive results for $\mathcal{BL}_1 L_q^{\geq n}$ when $q > 2$ and $n \geq 3$. The theorem below shows that no analogous positive theorem is possible for $1 < q \leq 2$, even with any fixed fiber dimension lower bound $n \geq 2$.

Idea of the proof

For $1 < q \leq 2$, the finite-dimensional space ℓ_q^n embeds isometrically into an L_1-space by q-stable random variables. Hence $L_1(\ell_q^n)$ and L_1 are mutually finitely representable: L_1 is a one-coordinate subspace of $L_1(\ell_q^n)$, while $L_1(\ell_q^n)$ embeds into L_1 after applying the stable embedding fiberwise. Heinrich's ultrapower theorem then gives isometric ultrapowers of the two spaces.

If the class $\mathcal{BL}_1 L_q^{\geq n}$ were axiomatizable after forgetting the lattice structure, it would be closed under ultrapowers and ultraroots. Since $L_1(\ell_q^n)$ belongs to the class, the ultrapower equivalence would force L_1 itself to belong to the class. But this is impossible: the dual of a nonzero $\mathcal{BL}_1L_q^{\geq n}$-space contains a 1-complemented copy of $\ell_{q'}^n$, while $L_1^* = L_\infty$ is 1-injective and cannot contain a 1-complemented non-injective $\ell_{q'}^n$.

Lemma 1 (stable finite-dimensional embedding)**.** *Let $1 < q \leq 2$ and $n \geq 2$. Then ℓ_q^n embeds linearly isometrically into an L_1-space.*

Proof. Let $\xi_1, \dots, \xi_n$ be independent symmetric q-stable random variables normalized so that $\mathbb{E}|\xi_1| = 1$. This first moment is finite because $q > 1$. By stability,

$$\sum_{j=1}^{n} a_j \xi_j$$

has the same distribution as

$$\left(\sum_{j=1}^{n} |a_j|^q \right)^{1/q} \xi_1.$$

Therefore

$$\mathbb{E} \left| \sum_{j=1}^{n} a_j \xi_j \right| = \left(\sum_{j=1}^{n} |a_j|^q \right)^{1/q}.$$

The map $a = (a_j) \mapsto \sum_j a_j \xi_j$ is the desired isometric embedding of ℓ_q^n into L_1. For $q = 2$, this is the Gaussian special case. $\square$

Lemma 2. *Let $1 < q < \infty$ and $n \geq 2$. The Banach space $L_1[0,1]$ is not linearly isometric to a member of $\mathcal{BL}_1L_q^{\geq n}$.*

Proof. We use a standard Banach-space invariant. The dual of an L_1-space is an L_∞-space and is 1-injective. Moreover, every 1-complemented subspace of a 1-injective space is again 1-injective.

By contrast, let Y be a nonzero member of $\mathcal{BL}_1L_q^{\geq n}$, represented as a band in a mixed-norm lattice $L_1(\Omega; L_q(\Omega_\omega))$. The condition $\geq n$ says that every nonzero base-band contains n disjoint fiber directions with the same base support, spanning a lattice copy of ℓ_q^n. Passing to the dual representation gives a contractive coordinate projection from Y^* onto the corresponding n-dimensional fiber dual $\ell_{q'}^n$. Thus Y^* contains a 1-complemented subspace isometric to $\ell_{q'}^n$, where $1/q + 1/q' = 1$.

For $1 < q < \infty$, the finite-dimensional space $\ell_{q'}^n$ is not 1-injective unless $q' = \infty$, equivalently $q = 1$. Since here $q > 1$, this is impossible in the dual of an L_1-space. Therefore $L_1[0,1]$ cannot be linearly isometric to a member of $\mathcal{BL}_1L_q^{\geq n}$. $\square$

Theorem 1 (negative fiber-dimension result)**.** *Let $1 < q \leq 2$ and $n \geq 2$. The class of Banach spaces linearly isometric to members of $\mathcal{BL}_1L_q^{\geq n}$ is not axiomatizable in the Banach-space language.*

Proof. Put

$$X = L_1[0,1], \qquad Y = L_1([0,1]; \ell_q^n).$$

The space Y is a member of $\mathcal{BL}_1L_q^{\geq n}$: it is the mixed-norm space with constant n-dimensional fiber ℓ_q^n.

The spaces X and Y are finitely equivalent. Indeed, X embeds isometrically into Y as the first coordinate. Conversely, by the stable embedding lemma there is an isometry $J : \ell_q^n \to L_1[0,1]$. Applying J fiberwise gives an isometric embedding

$$L_1([0,1]; \ell_q^n) \longrightarrow L_1([0,1] \times [0,1]),$$

and the latter is linearly isometric to $L_1[0,1]$ as a Banach space. Thus each space is finitely representable in the other with constant one.

By Heinrich's ultrapower characterization of finite equivalence, there are ultrafilters $\mathcal{U}, \mathcal{V}$ such that

$$X_{\mathcal{U}} \cong Y_{\mathcal{V}}$$

linearly isometrically. Suppose, for contradiction, that the Banach-space class generated by $\mathcal{BL}_1 L_q^{\geq n}$ were axiomatizable. Such a class is closed under ultrapowers and ultraroots. Since Y belongs to the class, $Y_{\mathcal{V}}$ belongs to the class. But $Y_{\mathcal{V}} \cong X_{\mathcal{U}}$, so X is an ultraroot of a member of the class. Closure under ultraroots would imply X itself belongs to the class, contradicting the preceding lemma. $\square$

Limitations

The theorem does not settle Raynaud's Question 6.17 for the unrestricted class $\mathcal{BL}_1 L_q$. The obstruction uses $L_1[0,1]$, which is itself in $\mathcal{BL}_1 L_q$ through a one-dimensional fiber representation. Thus it only rules out axiomatizability of the stronger subclasses that require every nonzero base-band to have fiber dimension at least $n \geq 2$.

The adjacent Nakano question from the same paper also remains open in the crossing-Hilbert-exponent range:

> Let us finish this section with mentioning an open question concerning classes of Nakano spaces.
>
> *Question* 5.8. Is the class $\mathcal{N}^B_{r,s}$ of Nakano (Banach) spaces associated with any variable exponent $p(\cdot)$ with values in $[r,s]$ axiomatizable, for every $1 \leq r < s < \infty$?
>
> 6. Other classes with DPIU

Novelty check

Local indexes were searched for arXiv:1603.07510, axiomatizable mixed-norm classes, $\mathcal{BL}_1 L_q$, Nakano spaces, ultraproducts, and finite representability. No existing packet for this source or this negative fiber-dimension result was found. Web and arXiv searches for the exact phrases "BL_1L_q axiomatizable", "L_1(L_q) axiomatizable", and "Nakano spaces axiomatizable" found the source paper but no later paper answering this variant.

Human review recommendation

Review as a partial negative result adjacent to Question 6.17. The main proof points to check are the exclusion lemma for $L_1[0,1]$, especially the assertion that nonzero $\mathcal{BL}_1 L_q^{\geq n}$-members have duals with a 1-complemented $\ell^n_{q'}$, and the use of Heinrich's finite-equivalence to ultrapower-isometry theorem.

References

[1] Y. Raynaud, *New axiomatizable classes of Banach spaces via disjointness-preserving isometries*, arXiv:1603.07510.

[2] S. Heinrich, *Ultraproducts in Banach spaces theory*, J. Reine Angew. Math. **313** (1980), 72–104.

[3] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces II: Function Spaces*, Springer, 1979.